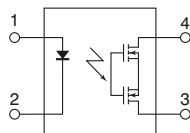




CAD Data

mm inch



FEATURES

1. Greatly increased load current in miniature SOP4-pin package (1.25A high capacity type added).
2. Greatly improved specifications allow you to use this in place of mercury and mechanical relays.

TYPICAL APPLICATIONS

- Measuring instruments
- Security and disaster-preventing system: use in I/O for alarm and security devices, etc.

PhotoMOS

TYPES

	Output rating*		Package	Part No.			Packing quantity	
	Load voltage	Load current		Tube packing style	Tape and reel packing style		Tube	Tape and reel
					Picked from the 1/2-pin side	Picked from the 3/4-pin side		
AC/DC dual use	60V	1.0A	SOP4-pin	AQY212GS	AQY212GSX	AQY212GSZ	1 tube contains: 100 pcs. 1 batch contains: 2,000 pcs.	1,000 pcs.
		1.25A		AQY212G2S	AQY212G2SX	AQY212G2SZ		

* Indicate the peak AC and DC values.

- Note: 1. For space reasons, the three initial letters of the part number "AQY", the surface mount terminal shape indicator "S" and the packing style indicator "X" or "Z" are not marked on the relay. (Ex. the label for product number AQY212G2SX is 212G2.)
2. For types with a built-in resistor, see page 92.

RATING

1. Absolute maximum ratings (Ambient temperature: 25°C 77°F)

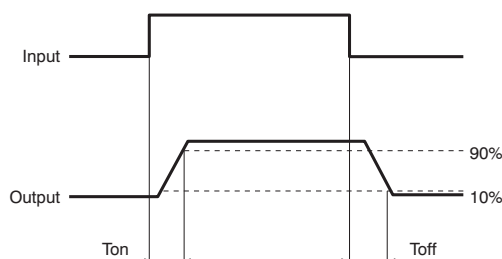
Item		Symbol	AQY212GS	AQY212G2S	Remarks
Input	LED forward current	I_F	50 mA		
	LED reverse voltage	V_R	5 V		
	Peak forward current	I_{FP}	1 A		$f = 100$ Hz, Duty factor = 0.1%
	Power dissipation	P_{in}	75 mW		
Output	Load voltage (peak AC)	V_L	60 V		
	Continuous load current	I_L	1.0 A	1.25 A	Peak AC, DC
	Peak load current	I_{peak}	3 A		100ms (1 shot), $V_L = DC$
	Power dissipation	P_{out}	300 mW		
Total power dissipation		P_T	350 mW		
I/O isolation voltage		V_{iso}	1,500 V AC		
Temperature limits	Operating	T_{opr}	-40°C to +85°C -40°F to +185°F		Non-condensing at low temperatures
	Storage	T_{stg}	-40°C to +100°C -40°F to +212°F		

GU SOP 1 Form A High Capacity (AQY212GS, AQY212G2S)

2. Electrical characteristics (Ambient temperature: 25°C 77°F)

Item			Symbol	AQY212GS	AQY212G2S	Condition
Input	LED operate current	Typical	I _{Fon}	1.1 mA		I _L = 100mA
		Maximum		3 mA		
	LED turn off current	Minimum	I _{Foff}	0.3 mA		I _L = 100mA
		Typical		1.0 mA		
	LED dropout voltage	Typical	V _F	1.32 V (1.14 V at I _F = 5 mA)		I _F = 50 mA
		Maximum		1.5 V		
Output	On resistance	Typical	R _{on}	0.34 Ω	0.2 Ω	I _F = 5 mA
		Maximum		0.7 Ω	0.5 Ω	I _L = Max. Within 1 s on time
	Off state leakage current	Maximum	I _{Leak}	1 μA		I _F = 0 mA V _L = Max.
Transfer characteristics	Turn on time*	Typical	T _{on}	1.3 ms		I _F = 5 mA
		Maximum		5.0 ms		I _L = 100 mA V _L = 10 V
	Turn off time*	Typical	T _{off}	0.1 ms		I _F = 5 mA
		Maximum		0.5 ms		I _L = 100 mA V _L = 10 V
	I/O capacitance	Typical	C _{iso}	0.8 pF		f = 1 MHz
		Maximum		1.5 pF		V _B = 0 V
	Initial I/O isolation resistance	Minimum	R _{iso}	1,000 MΩ		500 V DC
Max. switching frequency	Maximum	—	—	5 times/s	I _F = 5 mA duty = 50% V _L × I _L = 75 V·A	

*Turn on/Turn off time



RECOMMENDED OPERATING CONDITIONS

Please obey the following conditions to ensure proper relay operation and resetting.

Item	Symbol	Recommended value	Unit
Input LED current	I_F	5 to 10	mA

■ For Dimensions, see page 62.

■ For Schematic and Wiring Diagrams, see page 64.

■ For Cautions for Use, see page 71.

■ These products are not designed for automotive use.

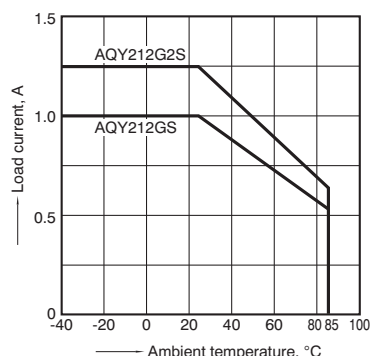
If you are considering to use these products for automotive applications, please contact your local Panasonic Electric Works technical representative.

For more information, see page 80.

REFERENCE DATA

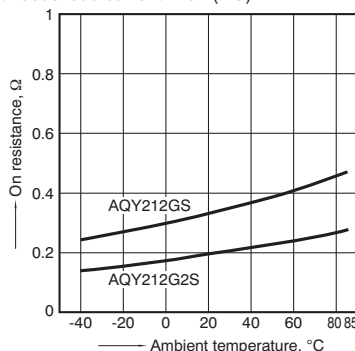
1. Load current vs. ambient temperature characteristics

Allowable ambient temperature: -40°C to $+85^{\circ}\text{C}$
 -40°F to $+185^{\circ}\text{F}$



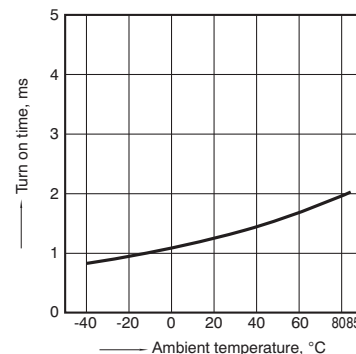
2. On resistance vs. ambient temperature characteristics

Measured portion: between terminals 3 and 4;
 LED current: 5 mA; Load voltage: Max. (DC)
 Continuous load current: Max. (DC)



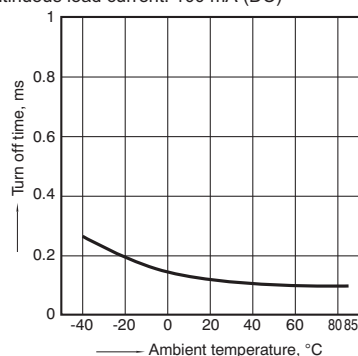
3. Turn on time vs. ambient temperature characteristics

LED current: 5 mA; Load voltage: 10 V (DC);
 Continuous load current: 100 mA (DC)



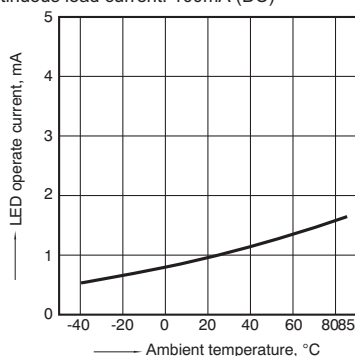
4. Turn off time vs. ambient temperature characteristics

LED current: 5 mA; Load voltage: 10 V (DC);
 Continuous load current: 100 mA (DC)



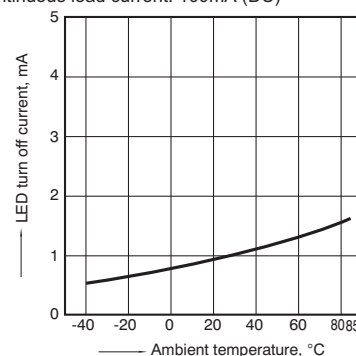
5. LED operate current vs. ambient temperature characteristics

Load voltage: 10 V (DC);
 Continuous load current: 100 mA (DC)



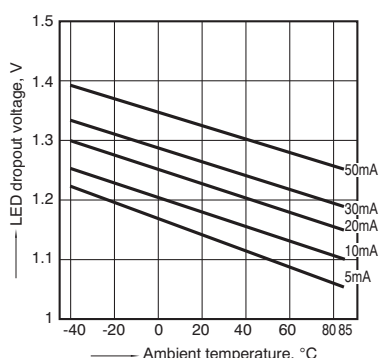
6. LED turn off current vs. ambient temperature characteristics

Load voltage: 10 V (DC);
 Continuous load current: 100 mA (DC)



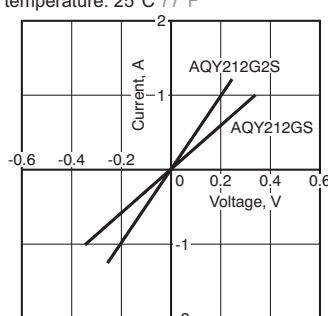
7. LED dropout voltage vs. ambient temperature characteristics

LED current: 5 to 50 mA



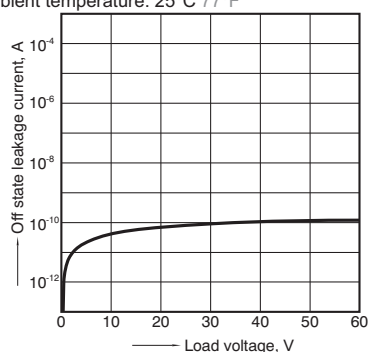
8. Current vs. voltage characteristics of output at MOS portion

Measured portion: between terminals 3 and 4;
 Ambient temperature: 25°C 77°F



9. Off state leakage current vs. load voltage characteristics

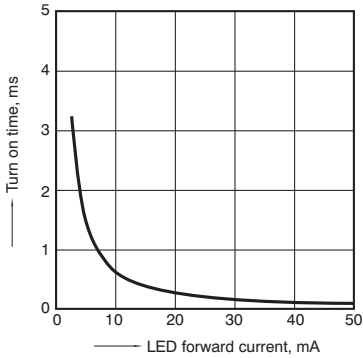
Measured portion: between terminals 3 and 4;
 Ambient temperature: 25°C 77°F



GU SOP 1 Form A High Capacity (AQY212GS, AQY212G2S)

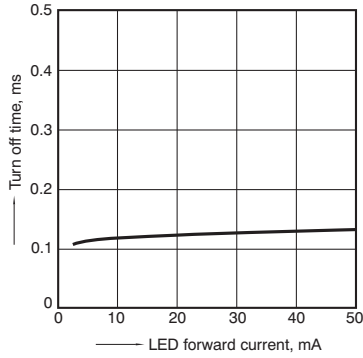
10. Turn on time vs. LED forward current characteristics

Measured portion: between terminals 3 and 4;
Load voltage: 10 V (DC);
Continuous load current: 100 mA (DC);
Ambient temperature: 25°C 77°F



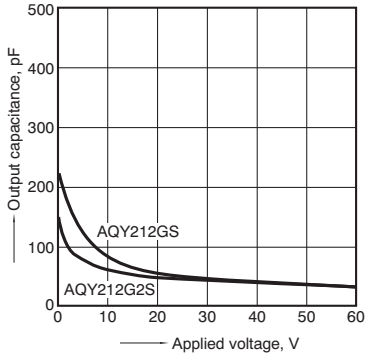
11. Turn off time vs. LED forward current characteristics

Measured portion: between terminals 3 and 4;
Load voltage: 10 V (DC);
Continuous load current: 100 mA (DC);
Ambient temperature: 25°C 77°F



12. Output capacitance vs. applied voltage characteristics

Measured portion: between terminals 3 and 4;
Frequency: 1 MHz;
Ambient temperature: 25°C 77°F



13. Max. switching frequency vs. load voltage and load current

LED current: 5 mA
Ambient temperature: 25°C 77°F

